Living Kolam

The doorstep kolam that dances back.

Concept by Sreya M Nambiar | DJXHack 2026, Everyday Magic

THE IDEA

A kolam made of light is projected onto the ground at a doorway or lobby. Step onto it and the glowing rice-flour dots scatter around your feet. Walk, and loops of light draw themselves around the dots in your path, in the traditional way. When a second person joins, the pattern grows into a full, symmetrical kolam. As the day brightens, it softly dissolves, and tomorrow begins fresh.

THE EVERYDAY MOMENT

Every dawn, in millions of homes, someone bends down at the threshold and draws a kolam with rice flour: a welcome, a blessing, a quiet daily ritual. It is washed away or walked over by evening, and drawn again the next morning. That impermanence is the heart of it. Living Kolam keeps the ritual and lets the whole family take part.

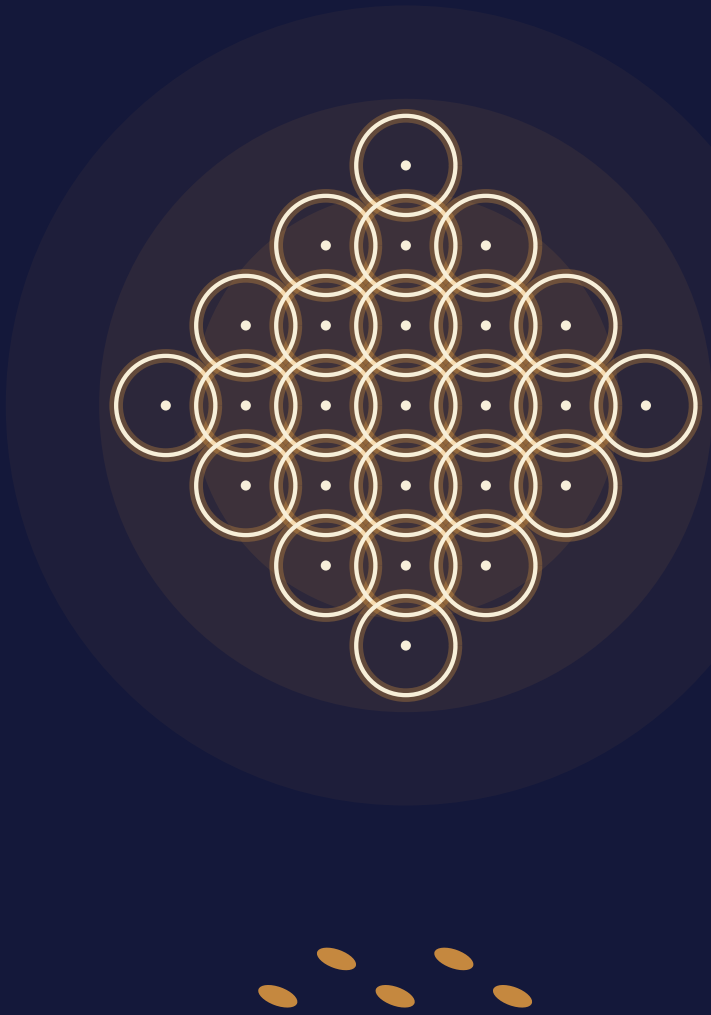
WHO IT IS FOR

Children like Sreya, who watch Paati draw the kolam and want to join but cannot yet draw the lines. **Grandparents** who can no longer bend down every morning. **Apartment families** with no mud doorstep to draw on. Anyone far from home who misses the ritual.

What they should feel: the warmth of a morning welcome. **What they should remember:** that a tradition can be played with, shared across ages, and still be gone gently by evening.

THE DOORSTEP, AT DAWN

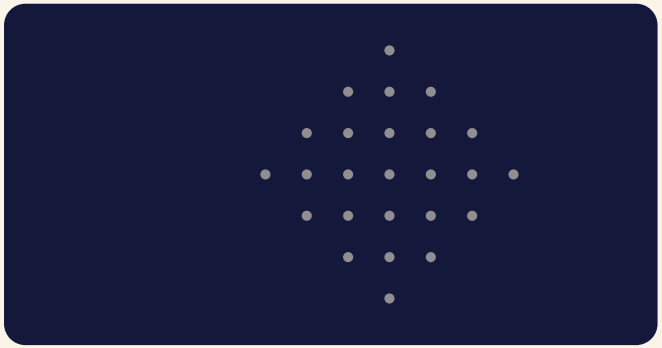
Light on the ground that answers your footsteps.



Concept illustration, drawn in code. Not a real capture.

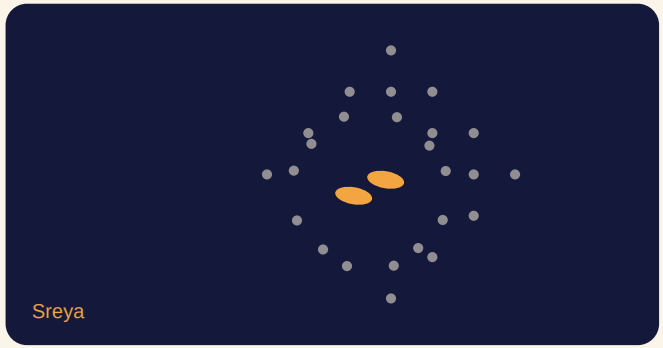
The interaction

What someone does, and how the kolam responds. Top-down view of the ground.



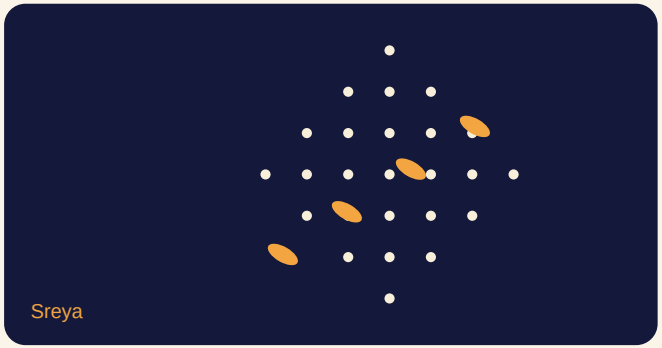
1 Dawn

An empty threshold. Faint dots glow on the ground, waiting like unwritten rice-flour dots.



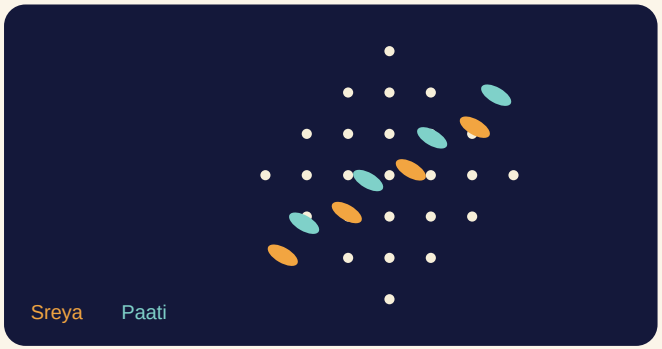
2 First step

Sreya steps in. The dots near her feet scatter outward, like flour disturbed by a toe.



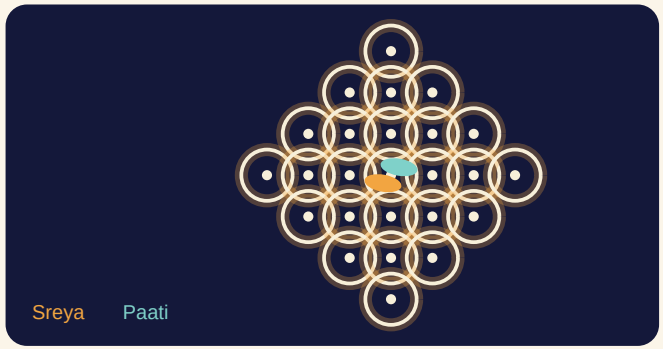
3 Walk

As she walks, loops of light draw themselves around the dots she passes. Her path becomes the pattern.



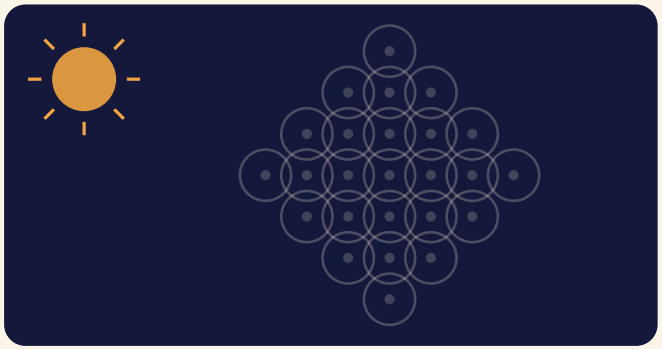
4 Together

Paati steps onto the opposite side. Her loops mirror Sreya's, and the kolam becomes symmetrical.



5 Complete

Both stand still. The full kolam glows and a soft temple-bell chime plays. Standing together is the finish.



6 Sunrise

Daylight rises and the kolam dissolves like it was swept away. The ground is clear for tomorrow.

ACTION > RESPONSE

Step

Dots scatter around the foot, then settle.

Walk

Light loops draw along the walker's path.

Stand together

Two people complete the symmetry; chime plays.

Feel / remember: a welcome that responds to you, and that the ritual is better with two generations on the same ground.

Possible approach

An exploratory plan. A finished prototype is not required, and none is claimed here.

SENSE
Ceiling camera or depth sensor (LiDAR / Kinect) sees feet on the floor

→

TRACK
OpenCV / MediaPipe finds each person's position and path

→

GENERATE
Kolam engine: pulli dot grid, loop rules, mirror symmetry (p5.js / TouchDesigner)

→

PROJECT
Short-throw projector draws light on the floor; speaker plays soft chime

TOOLS AND MATERIALS

Real installation: overhead depth camera, short-throw projector (floor-facing), small speaker, a mini PC, and a matte floor mat so the light reads clearly at a doorway or lobby.

Kolam logic: a pulli (dot) grid; each dot gets a loop only when someone's path comes near it. Two walkers trigger mirror symmetry. Loops fade as ambient light rises.

Low-cost version: a phone or laptop webcam and a browser canvas. The ground becomes the screen, and your finger or feet become the input. This can also serve as the optional web demo.

WHAT IS REAL vs SIMULATED

Concept video	Simulated: acted / AI-generated scenes of the projection
Pages 1 and 2 visuals	Illustrations drawn in code; not captured from a build
Web demo (optional)	Browser canvas with touch or webcam, if linked in the form
Full floor projection	Proposed; not built yet

WHERE IT COULD LIVE

Apartment lobbies, temple streets, Pongal and Margazhi festivals, school entrances, cultural centres. The same engine can adopt other regional floor traditions such as muggu and rangoli.

CREDITS

Concept, idea and interaction design: Sreya M Nambiar. Illustrations on pages 1 and 2 are drawn in code.